CLASS 9 · CBSE · GANITA MANJARI PART 1 · DPP SHEET

Chapter 4: Exploring Algebraic Identities

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Expand $(5m + 2n)^2$.
2. Find 103^2 using an identity.
3. Find 96×104 using an identity.
4. Factorise $25x^2 + 40xy + 16y^2$.
5. Factorise $x^2 - 7x + 10$.
6. Factorise $2x^2 + 7x + 6$.
7. Expand $(x + 2y + 3z)^2$.
8. Expand $(3p - 2)^3$.
9. Factorise $125m^3 - 8$.
10. Simplify $\frac{x^2 + 5x + 6}{x^2 - 4}$, assuming the denominator is not zero.

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Answer Key

1. $25m^2 + 20mn + 4n^2$
2. $(100 + 3)^2 = 10609$
3. $100^2 - 4^2 = 9984$
4. $(5x + 4y)^2$
5. $(x - 2)(x - 5)$
6. $(2x + 3)(x + 2)$
7. $x^2 + 4y^2 + 9z^2 + 4xy + 12yz + 6zx$
8. $27p^3 - 54p^2 + 36p - 8$
9. $(5m - 2)(25m^2 + 10m + 4)$
10. $\frac{(x + 2)(x + 3)}{(x + 2)(x - 2)} = \frac{x + 3}{x - 2}$

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